

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

303000097 - Sistemas Dinámicos Avanzados

DEGREE PROGRAMME

30AH - Máster Universitario En Matemáticas Avanzadas

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

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1. Description

1.1. Subject details

Name of the subject	303000097 - Sistemas Dinámicos Avanzados
No of credits	6 ECTS
Type	Optional/elective
Academic year of the programme	First year
Semester of tuition	Semester 1
Tuition period	September-January
Tuition languages	English
Degree programme	30AH - Máster Universitario en Matemáticas Avanzadas
Centre	30 - Esc. Politéc. Enseñanza Superior (epes)
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Ramon Barral Lijo (Subject coordinator)		ramon.barral@upm.es	- -

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

3.2. Other recommended learning outcomes

- Linear algebra
- Undergraduate analysis
- Undergraduate topology

4. Skills and learning outcomes *

4.1. Skills to be learned

CO1 - Conocer los principales métodos y teorías matemáticas, próximos a la frontera del conocimiento, que permitan tener una amplia perspectiva en al menos tres de las siguientes áreas: Análisis Matemático, Métodos Numéricos, Ecuaciones Diferenciales, Estadística y Ciencia de Datos, Álgebra, Geometría y Topología. TIPO: Conocimientos o contenidos.

CO2 - Conocer demostraciones matemáticas avanzadas que permitan relacionar de una manera rigurosa las hipótesis con las conclusiones en al menos tres de las áreas relacionadas en el CO1. TIPO: Conocimientos o contenidos.

CO3 - Conocer diversas aplicaciones de las teorías matemáticas y su modelización en otros entornos o disciplinas que permitan adaptar los conocimientos matemáticos a nuevos problemas. TIPO: Conocimientos o contenidos.

CP2 - Ser capaz de trabajar en equipos multidisciplinares que involucren conocimientos matemáticos avanzados en áreas teóricas o aplicadas. TIPO: Competencias.

CP3 - Seleccionar y comprender la bibliografía científica sobre un tema concreto, extrayendo la información relevante. TIPO: Competencias.

CP4 - Comunicar de forma oral y escrita conocimientos y conclusiones científicas, valorando diferentes interpretaciones en públicos especializados y no especializados de una manera clara y rigurosa. TIPO: Competencias

CP5 - Complementar la formación matemática de manera autónoma en avances específicos de las Matemáticas, seleccionando la bibliografía adecuada y planteando cuestiones de interés. TIPO: Competencias

HA1 - Aplicar los conocimientos matemáticos adquiridos a la resolución de problemas nuevos de una manera crítica, generando hipótesis de trabajo claras, definiciones precisas, nuevos enfoques con fundamento matemático y soluciones originales. TIPO: Habilidades o destrezas

HA2 - Ser capaz de abstraer y detectar las dificultades relevantes ante problemas complejos para poder plantearlos correctamente y resolverlos con técnicas matemáticas avanzadas. TIPO: Habilidades o destrezas

HA3 - Dominar el rigor matemático en las técnicas de demostración aprendidas para su posterior uso en el enunciado y demostraciones de nuevos teoremas. TIPO: Habilidades o destrezas

4.2. Learning outcomes

RA2 - Los resultados de aprendizaje en las tres categorías propuestas: conocimientos/contenidos, habilidades y competencias son los señalados en la memoria del título

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

Dynamical systems form a central area of modern mathematics concerned with the evolution of systems over time under deterministic rules. Originating in classical mechanics with the work of Newton and Poincaré, the theory has developed into a rich and diverse field, offering powerful tools for understanding stability, bifurcations, long-term behavior, and the emergence of chaotic dynamics. Its influence extends well beyond mathematics, playing a fundamental role in physics, biology, economics, and data science, where complex temporal phenomena are ubiquitous.

At its core, the study of dynamical systems seeks to answer deep and often subtle questions: How do small changes in initial conditions affect future states? What structures or patterns persist under iteration? When does randomness emerge from deterministic rules? To tackle such questions, the field brings together geometry, topology, analysis, and probability.

This master-level course provides an introduction to advanced topics in dynamical systems, emphasizing both topological and measure-theoretic approaches. Without aiming to be exhaustive, the course will explore key concepts such as transitivity, chaos, invariant measures, and entropy, through concrete examples and fundamental theorems. The goal is to equip students with a solid theoretical foundation and a global perspective on the qualitative behavior of deterministic systems, preparing them for further research or interdisciplinary applications.

5.2. Syllabus

1. Topological Dynamics and Foundational Concepts

- 1.1. Contractions and linear maps
- 1.2. Topological transitivity and minimality
- 1.3. The logistic map and symbolic dynamics
- 1.4. Devaney chaos

2. Attractors and hyperbolicity

- 2.1. Attractors
- 2.2. The dyadic solenoid
- 2.3. Hyperbolicity
- 2.4. Bifurcations

3. Ergodic theory and measurable dynamics

- 3.1. Invariant measures

- 3.2. Bernoulli shift
- 3.3. Ergodicity and mixing
- 3.4. Krylov-Bogoliubov and Birkhoff theorems
- 3.5. Equidistribution
- 3.6. Measure-theoretic entropy

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Lecture Duration: 02:00 Lecture Lecture Duration: 02:00 Lecture			Reading seminar - Pairs of students will be asked to prepare material for a lecture and teach said materials in class in front of their classmates. Group presentation Progressive assessment Presential Duration: 00:00
2	Lecture Duration: 02:00 Lecture Lecture Duration: 02:00 Lecture			
3	Lecture Duration: 02:00 Lecture Lecture Duration: 02:00 Lecture			
4	Lecture Duration: 02:00 Lecture Lecture Duration: 02:00 Lecture			
5	Lecture Duration: 02:00 Lecture Lecture Duration: 02:00 Lecture			
6	Lecture Duration: 02:00 Lecture Lecture Duration: 02:00 Lecture			

7	Lecture Duration: 02:00 Lecture Lecture Duration: 02:00 Lecture			
8	Lecture Duration: 02:00 Lecture Lecture Duration: 02:00 Lecture			
9	Lecture Duration: 02:00 Lecture Lecture Duration: 02:00 Lecture			
10	Lecture Duration: 02:00 Lecture Lecture Duration: 02:00 Lecture			
11	Lecture Duration: 02:00 Lecture Lecture Duration: 02:00 Lecture			
12	Lecture Duration: 02:00 Lecture Lecture Duration: 02:00 Lecture			
13	Lecture Duration: 02:00 Lecture Lecture Duration: 02:00 Lecture			
14	Lecture Duration: 02:00 Lecture Lecture Duration: 02:00 Lecture			

15	Lecture Duration: 02:00 Lecture Lecture Duration: 02:00 Lecture			
16				
17				Final exam - Problems Written test Global examination Presential Duration: 01:00 Final exam - Theory Written test Global examination Presential Duration: 01:00 Notes - Students must submit the handwritten notes used in their class presentation. Group work Progressive assessment Presential Duration: 00:00 Exam - Theory Written test Progressive assessment Presential Duration: 01:00 Exam - Problems Written test Progressive assessment Presential Duration: 01:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
1	Reading seminar - Pairs of students will be asked to prepare material for a lecture and teach said materials in class in front of their classmates.	Group presentation	Face-to-face	00:00	30%	0 / 10	
17	Notes - Students must submit the handwritten notes used in their class presentation.	Group work	Face-to-face	00:00	20%	0 / 10	CP3 HA3 HA2 CP5 CP4 CO2 CO3 HA1 CO1 CP2
17	Exam - Theory	Written test	Face-to-face	01:00	30%	/ 10	CP3 HA3 HA2 CP5 CP4 CO2 CO3 HA1 CO1 CP2
17	Exam - Problems	Written test	Face-to-face	01:00	20%	/ 10	CP3 HA3 HA2 CP5 CP4 CO2 CO3 HA1 CO1 CP2

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Final exam - Problems	Written test	Face-to-face	01:00	60%	0 / 10	CP3 HA3 HA2 CP5 CP4 CO2 CO3 HA1 CO1 CP2
17	Final exam - Theory	Written test	Face-to-face	01:00	40%	0 / 10	CP3 HA3 HA2 CP5 CP4 CO2 CO3 HA1 CO1 CP2

7.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Exam - Theory	Written test	Face-to-face	01:00	60%	/ 10	CP3 HA3 HA2 CP5 CP4 CO2 CO3 HA1 CO1 CP2

Exam - Problems	Written test	Face-to-face	01:00	40%	/ 10	HA3 HA2 CP5 CP4 CO2 CP3 CO3 HA1 CO1 CP2
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7.2. Assessment criteria

Students opting for continuous evaluation will be asked to work in pairs and prepare a topic to present to their classmates. They must also submit the handwritten notes they plan to follow during the presentation. Their performance during the presentation will be worth three points, and the handwritten notes will be worth two points. At the end of the course, there will be an exam: the theory questions will be worth three points, and the problem-solving questions will be worth two points.

Students opting for a global evaluation will take a single exam at the end of the course. In this exam, the theory questions will be worth six points, and the problem-solving questions will be worth four points.

In July, there will be a resit exam. In this exam, the theory questions will be worth six points, and the problem-solving questions will be worth four points.

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
R.L.Devaney, "An Introduction to Chaotic Dynamical Systems" (3rd ed.), CRC Press, 2022.	Bibliography	
M.W.Hirsch, S.Smale and R.L.Devaney, "Differential Equations, Dynamical Systems, and an Introduction to Chaos" (3 ^a ed.), Elsevier, 2013.	Bibliography	
A.Katok and B.Hasselblatt, "Introduction to the Modern Theory of Dynamical Systems", Encyclopedia Math.Appl., vol. 54,	Bibliography	Cambridge University Press, 1995.
J. Moser and E.J. Zehnder, "Notes on Dynamical Systems", Courant Lect. Notes Math., vol.12, copublished by AMS and the Courant Institute of Mathematical Sciences, 2005.	Bibliography	
C. Robinson, "Dynamical Systems: Stability, Symbolic Dynamics and Chaos", Advanced Studies in Mathematics, CRC Press (1995).	Bibliography	
S. Strogatz, "Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry and Engineering" (2nd ed.), Studies in Nonlinearity, Westview Press (2015).	Bibliography	